

Freight Rate Prediction — ML Assessment Executive Summary Predict posted_rate (USD) for freight loads. Final model: HistGradientBoosting with feature set Q, raw posted_rate target, and absolute_error loss. Trained on 48,000 Jan-Oct rows; inferred 12,000 Nov-Dec holdout rows. Development chronological validation MAE: 93.56 (primary) and 102.57 (sensitivity). Primary RMSE: 630.27. High-rate MAE: 816.54. Long-haul MAE: 164.93. These are NOT Nov-Dec holdout scores. score.py validates format only (exit code 0).

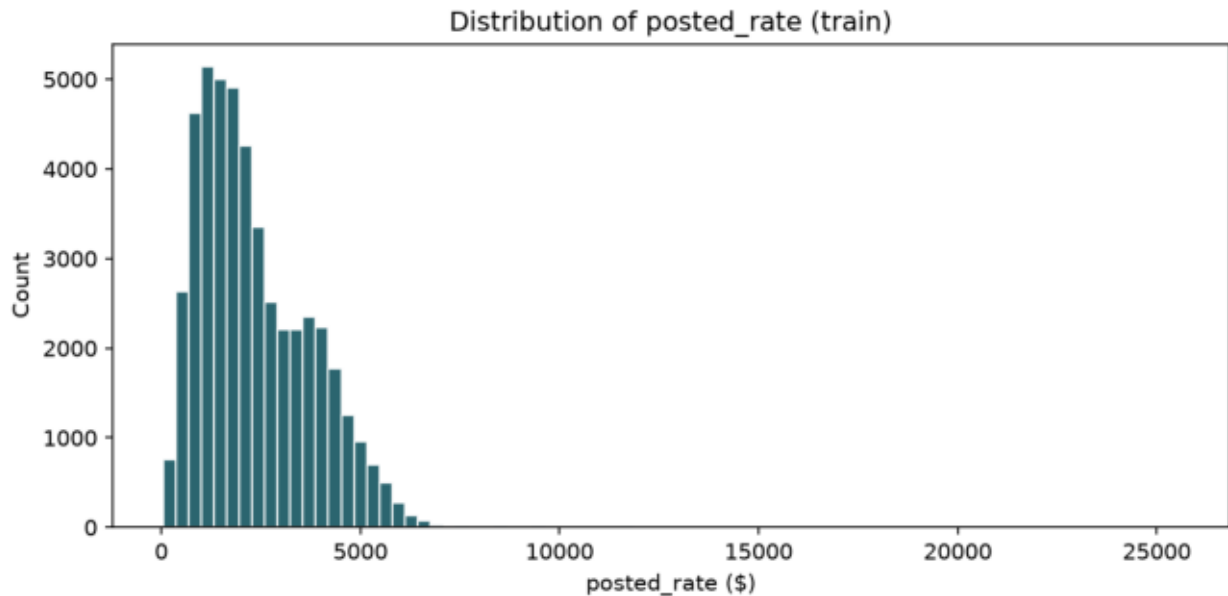
Dataset & Validation Strategy train-test.csv: 48,000 labeled rows. validation.csv: 12,000 unlabeled holdout. Primary split: Jan-Aug train → Sep-Oct validate. Sensitivity: Jan-Sep train → Oct validate. No random shuffle. Preprocessing fit on training fold only.

Feature Set Q & Final Model Features: distance, log_distance, distance_bin, weight, weight_is_missing, quote_signal, equipment. Excluded: raw market_index, route/city OHE, geographic extras. Target: raw posted_rate (no log1p/expm1). Loss: absolute_error. Hyperparameters: max_depth=6, l2=0.1, lr=0.08, max_iter=300, random_state=42.

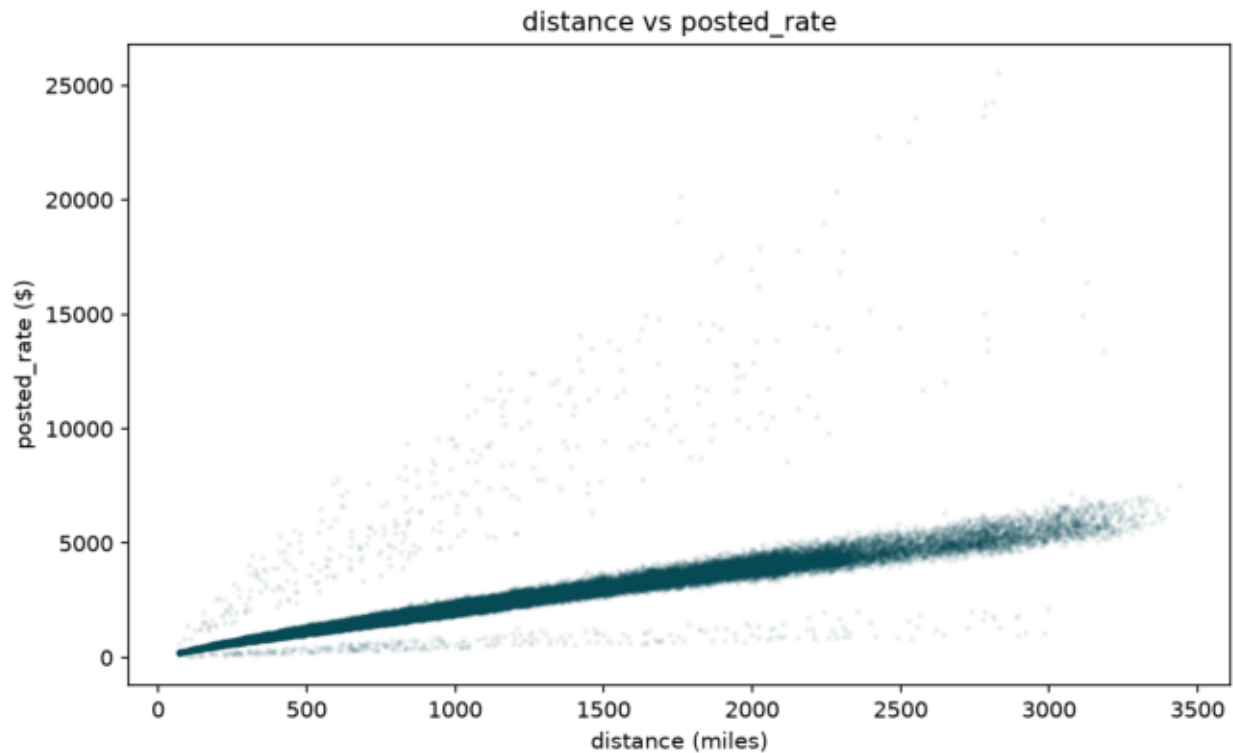
Model Comparison (development chronological validation) Global median: 1148.92 | Distance linear: 196.95 | Ridge: 148.06 HistGB FULL: 129.69 | HistGB log1p FULL: 118.83 HistGB Q log1p + squared_error (earlier candidate): 106.83 / 113.29 HistGB Q raw + absolute_error (FINAL): 93.56 primary / 102.57 sensitivity Final predictions: mean \$2,350.58, median \$2,015.78, min \$201.58, max \$6,652.40.

Limitations • No local Nov-Dec ground truth • score.py does not compute accuracy • Long-tail underprediction remains • December chart flat at \$815.51 (Q excludes calendar features)

posted_rate distribution

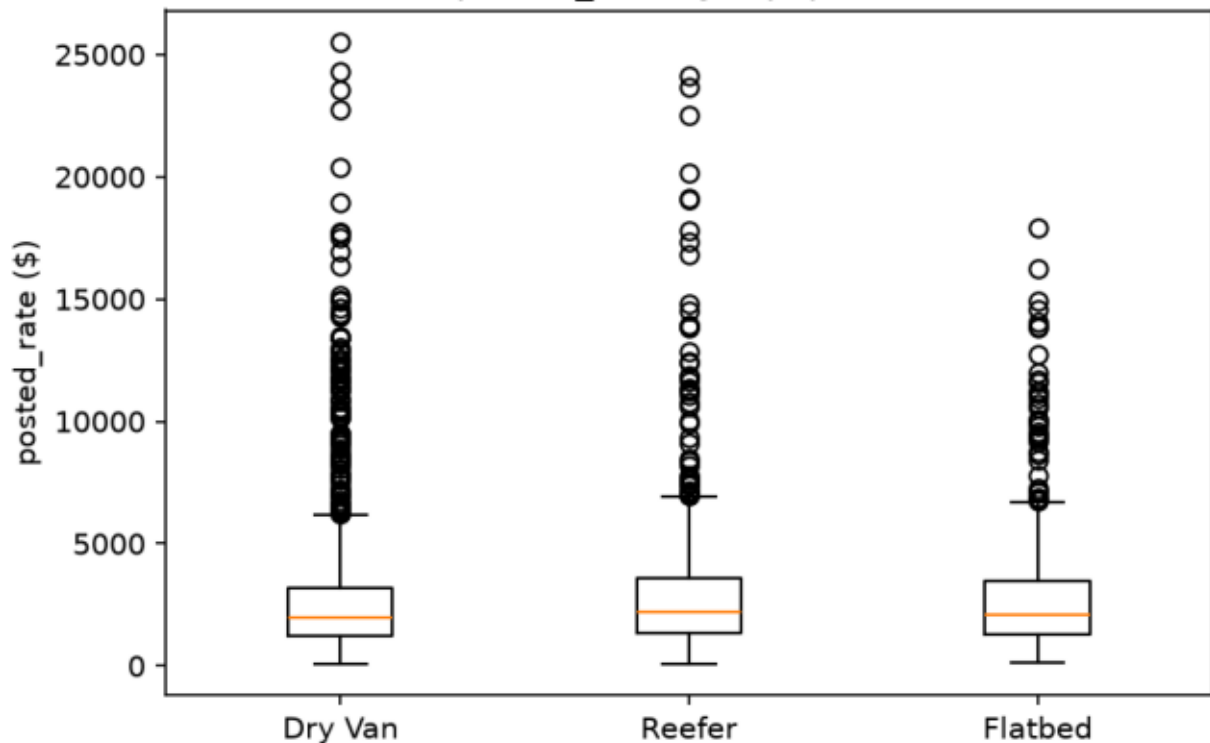


Distance vs posted_rate



Rate by equipment

posted_rate by Equipment



December 2025 candidate prediction

Candidate: December 2025 Predicted Load Rate

